Incentive Compatibility in a Discriminatory Limit-Order Auction

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Abstract

We study a symmetric discriminatory double auction in which n agents hold Gaussian private signals about an unknown asset value and submit report pairs $(\hat{\mu}_i, \hat{\sigma}_i)$ that parametrise their limit-order books. The mechanism fills orders at the reported limit prices (pay-as-bid/ask), and the market-clearing price is determined endogenously. Each agent believes the MCP concentrates around their own posterior mean at a rate proportional to their own uncertainty, yielding a signal-to-noise ratio $\kappa(n) = n/(n+1)$ that is common to all agents.

Our main results establish that truthful reporting of both the posterior mean and the posterior standard deviation can be sustained as an incentive-compatible equilibrium. Specifically, we show: (i) for any fixed reported radius $L > 0$, truth of the midpoint $\hat{\mu} = \mu_i$ is the unique maximiser of expected profit regardless of the shape of m (Proposition 3.3); (ii) the first-order condition for the optimal reported radius L^* reduces to a single zero-crossing condition on the marginal-profit density g_0 (Proposition 4.2); and (iii) truthful spread reporting is incentive compatible if and only if the depth profile m is strictly positive everywhere (Theorem 5.1), with the calibration condition reducing to a single transcendental equation in $\kappa(n)$ (Remark 5.4).

We then study market liquidity under endogenous entry. Because $\kappa(n) = n/(n+1)$, the optimal reported spread is proportional to the true spread for all agents, giving a clean separating equilibrium. There exists a unique market size $n^\dagger$ at which the mechanism is simultaneously truthful for all agents (Theorem 6.9), and a unique outside option level $c^\dagger$ that implements it as a rational expectations equilibrium (Theorem 6.8). As the pool of potential entrants grows without bound, the equilibrium spread narrows to zero and depth diverges (Theorem 6.11).

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1 Model

1.1 Private Signals and Reports

Let $n \in \mathbb{N}$ agents participate in the game, each endowed with budget $2Q > 0$. Each agent i receives a private signal (μ_i, σ_i), where

$$\mu_i \sim \mathcal{N}(\mu, \sigma^2)$$

for unknown parameters $(\mu, \sigma) \in \mathbb{R} \times \mathbb{R}_+$. The quantity μ_i is agent i's posterior estimate of the asset's true value μ, and $\sigma_i > 0$ measures the uncertainty of that estimate. Each agent submits a report $(\hat{\mu}_i, \hat{\sigma}_i) \in \mathbb{R} \times \mathbb{R}_+$.

1.2 Order Books and the Depth Profile

The mechanism enforces a common marginal depth function $m : \mathbb{R}_+ \to \mathbb{R}_+$ for all agents.

Definition 1.1. A report $(\hat{\mu}, \hat{\sigma})$ induces a *no-trade band* $[\hat{\mu} - \hat{\sigma},\ \hat{\mu} + \hat{\sigma}]$. At radius $r > \hat{\sigma}$, the agent posts buy (resp. sell) depth $m(r)$ at price $\hat{\mu} - r$ (resp. $\hat{\mu} + r$). The function $m : \mathbb{R}_+ \to \mathbb{R}_+$ is càdlàg, nonincreasing, and satisfies

$$\int_{\hat{\sigma}}^{\infty} m(r)\, dr = Q$$

The normalisation ensures that each agent's total posted quantity on each side equals the budget Q. The no-trade band captures the agent's indifference region: orders are only placed outside it, reflecting the belief that trades within it are unlikely to be profitable.

1.3 Discriminatory Auction and Market-Clearing Price

Define the cumulative buy and sell volumes above and below a price $p \in \mathbb{R}$ as

$$V^B(p) = \sum_{i=1}^{n} \int_{\max(\hat{\sigma}_i,\, \hat{\mu}_i - p)}^{\infty} m(r)\, dr$$
$$V^S(p) = \sum_{i=1}^{n} \int_{\max(\hat{\sigma}_i,\, p - \hat{\mu}_i)}^{\infty} m(r)\, dr$$

Definition 1.2. The traded volume at price p is $T(p) := \min\{V^B(p),\ V^S(p)\}$. The *market-clearing price* (MCP) is

$$p^* = \arg\max_{p \in \mathbb{R}} T(p)$$

with a midpoint tie-breaking rule. All buy orders at $p > p^*$ and sell orders at $p < p^*$ are executed at their respective limit prices (discriminatory/pay-as-bid–ask rule).

1.4 Beliefs about Other Agents and Auction Noise

A key modelling choice concerns how agent i forms beliefs about the location of p^*. Agent i's posterior is $\mu \mid \mathcal{F}_i \sim \mathcal{N}(\mu_i, \sigma_i^2)$, so they model the MCP as concentrating around their own μ_i. With n participants each contributing independent information, the residual variance of p^* around μ_i scales as σ_i^2/n: more participants sharpen price discovery by a factor proportional to the agent's own uncertainty.

Assumption 1.3. Agent i's conditional belief about the MCP satisfies

$$\mu \mid \mathcal{F}_i \sim \mathcal{N}(\mu_i,\, \sigma_i^2), \qquad p^* \mid \mathcal{F}_i \sim \mathcal{N}\left(\mu_i,\, \sigma_i^2 \cdot \frac{n+1}{n}\right)$$

where $\mathcal{F}_i = \sigma(\mu_i, \sigma_i)$ is the σ-algebra generated by agent i's private signal. Equivalently, writing $X := p^* - \mu$ for the auction noise and $Z := \mu - \mu_i$ for the signal error,

$$Z \sim \mathcal{N}(0, \sigma_i^2), \qquad X \sim \mathcal{N}\left(0, \frac{\sigma_i^2}{n}\right), \qquad X \perp\!\!\!\perp Z \mid \mathcal{F}_i$$

The independence of X and Z holds whenever the MCP is determined by the aggregate order book in a way that is orthogonal to any single agent's idiosyncratic signal error, which is natural when n is large. A central consequence of Assumption 1.3 is that the signal-to-noise ratio

$$\kappa(n) := \frac{\sigma_i^2}{\sigma_i^2 + \sigma_i^2/n} = \frac{n}{n+1} \tag{1}$$

is the same for all agents and depends only on market size n. In particular $\kappa(n)$ is strictly increasing in n with $\kappa(n) \to 1$ as $n \to \infty$.

Remark 1.4. As n increases, two effects operate simultaneously. First, more participants reduce noise: p^* concentrates more tightly around μ, improving price discovery. Second, competitive pressure increases: with more agents posting orders, the MCP is more strongly pulled by the aggregate book, shrinking the window of profitable execution for any individual. Because both effects scale identically with σ_i, they cancel and $\kappa(n) = n/(n+1)$ is agent-independent. The ratio κ is therefore a pure measure of market thickness.

2 Rational Agents and Expected Profit

2.1 Marginal Profit at a Given Radius

An agent profits whenever he buys below (or sells above) the true value μ. Fix agent i and suppress the subscript for readability. For a report $(\hat{\mu}, \hat{\sigma})$ with $L := \hat{\sigma}$, define

$$\boldsymbol{\theta} := (\Delta, Y) \qquad \Delta := \hat{\mu} - \mu \qquad Y := p^* - \hat{\mu}$$

so Δ is the signed reporting error and Y the displacement of the MCP from the reported midpoint.

Proposition 2.1. *For any $r \geq L$, the contribution to profit from the order at radius r is*

$$C(r \mid \boldsymbol{\theta}) = m(r)\Big((r - \Delta)\,\mathbf{1}_{\{Y \leq -r\}} + (\Delta + r)\,\mathbf{1}_{\{Y > r\}}\Big)$$

Proof. At radius $r \geq \hat{\sigma}$, the agent sells at $\hat{\mu} + r$ whenever $Y > r$, with profit per unit $(\hat{\mu} + r) - \mu = r + \Delta$. Symmetrically, the agent buys at $\hat{\mu} - r$ whenever $Y \leq -r$, with profit per unit $\mu - (\hat{\mu} - r) = r - \Delta$. Multiplying by $m(r)$ and summing gives the result. $\square$

2.2 Setup and Standing Assumptions

Notation 2.2. Fix agent i and suppress the subscript. In terms of Assumption 1.3,

$$\begin{aligned} Z &:= \mu - \mu_i \sim \mathcal{N}(0, \sigma_i^2), \\ X &:= p^* - \mu \sim \mathcal{N}(0, \sigma_i^2/n), \\ W &:= X + Z = p^* - \mu_i \sim \mathcal{N}\big(0,\, \sigma_i^2 \cdot \tfrac{n+1}{n}\big) \end{aligned}$$

For a report $(\hat{\mu}, L) = (\mu_i + a,\, L)$ with deviation $a := \hat{\mu} - \mu_i$,

$$\Delta = a - Z, \qquad Y = W - a$$

Remark 2.3. Since X and Z are independent centred Gaussians (Assumption 1.3), the pair (X, Z) is jointly centred Gaussian with $(X, Z) \overset{d}{=} (-X, -Z)$, hence $(W, Z) \overset{d}{=} (-W, -Z)$.

Assumption 2.4. The marginal depth function is positive and integrable

$$m(r) \geq 0 \quad \forall\, r \geq L \qquad \int_L^\infty m(r)\, dr < \infty$$

Integrating Proposition 2.1 over all radii gives the total profit:

$$\Pi_m(\boldsymbol{\theta}) = \int_L^\infty m(r) \Big((r - \Delta)\, \mathbf{1}_{\{Y \leq -r\}} + (\Delta + r)\, \mathbf{1}_{\{Y > r\}} \Big)\, dr \tag{2}$$

The ex-ante expected profit conditional on $\mathcal{F}_i$ is

$$J_m(a, L) := \mathbb{E}_{\boldsymbol{\theta}}[\Pi_m(\boldsymbol{\theta}(a, L)) \mid \mathcal{F}_i]$$

By Fubini's theorem (Theorem A.1), this decomposes as

$$J_m(a, L) = \int_L^\infty m(r)\, g_a(r)\, dr \tag{3}$$

where the *marginal profit density* at radius r is

$$g_a(r) := \mathbb{E}\Big[(r - \Delta)\, \mathbf{1}_{\{Y \leq -r\}} + (\Delta + r)\, \mathbf{1}_{\{Y > r\}} \Big] \tag{4}$$

$$= \mathbb{E}\Big[(r - a + Z)\, \mathbf{1}_{\{W \leq a - r\}} + (a - Z + r)\, \mathbf{1}_{\{W > a + r\}} \Big] \tag{5}$$

3 Optimality of Truthful Midpoint Reporting

For any fixed $L > 0$, we show that reporting the true posterior mean as the midpoint is the unique best response, regardless of the shape of m. The argument establishes that $J_m(a, L)$ is strictly concave in a with its unique stationary point at $a = 0$.

Lemma 3.1. *Fix $r > 0$. The map $a \mapsto g_a(r)$ is twice continuously differentiable with*

$$\frac{\partial g_a}{\partial a}(r) = \mathbb{P}(W > a + r) - \mathbb{P}(W \leq a - r) = 1 - F_W(a + r) - F_W(a - r) \tag{6}$$

$$\frac{\partial^2 g_a}{\partial a^2}(r) = -f_W(a + r) - f_W(a - r) \tag{7}$$

where F_W and f_W are the CDF and density of W.

Proof. Write $\phi(a; z, w, r) := (r - a + z)\,\mathbf{1}_{\{w \le a-r\}} + (a - z + r)\,\mathbf{1}_{\{w > a+r\}}$ so $g_a(r) = \mathbb{E}[\phi(a; Z, W, r)]$ by (5). For fixed (z, w, r), $a \mapsto \phi$ is piecewise affine with breakpoints at $a \in \{w - r, w + r\}$ and $|\partial_a \phi| \le 1$. Since W has a continuous density the breakpoints have probability zero, so dominated convergence (Theorem A.2) justifies differentiation under the expectation:

$$\frac{\partial g_a}{\partial a}(r) = \mathbb{E}\big[-\mathbf{1}_{\{W \le a-r\}} + \mathbf{1}_{\{W > a+r\}}\big] = \mathbb{P}(W > a + r) - \mathbb{P}(W \le a - r)$$

giving (6). Since W is Gaussian, $F_W \in C^\infty$; differentiating once more yields (7). □

Lemma 3.2. *For each $r > 0$, $a \mapsto g_a(r)$ is strictly concave with unique global maximum at $a = 0$.*

Proof. By Proposition A.10, $F_W(-x) = 1 - F_W(x)$. Evaluating (6) at $a = 0$:

$$\frac{\partial g_a}{\partial a}(r)\Big|_{a=0} = 1 - F_W(r) - F_W(-r) = 1 - F_W(r) - (1 - F_W(r)) = 0$$

so $a = 0$ is stationary. Since $f_W > 0$ on $\mathbb{R}$, equation (7) gives $\partial^2 g_a/\partial a^2 < 0$ everywhere, so $a \mapsto g_a(r)$ is strictly concave and $a = 0$ is its unique global maximiser. □

Proposition 3.3. *Let Assumptions 1.3 and 2.4 hold. For any fixed $L > 0$, $\sup_{a\in\mathbb{R}} J_m(a, L) = J_m(0, L)$ with the supremum attained uniquely at $a = 0$. Reporting the true posterior mean $\hat{\mu} = \mu_i$ is the unique best response for any choice of L.*

Proof. Gaussian tail bounds give $|g_a(r)| \le C(1 + r)$ uniformly in a, so dominated convergence (Theorem A.2) justifies differentiation under the integral in (3). Using (6),

$$\frac{\partial J_m}{\partial a}(a, L) = \int_L^\infty m(r)\Big(\mathbb{P}(W > a + r) - \mathbb{P}(W \le a - r)\Big)\,dr$$

At $a = 0$ each integrand vanishes by symmetry of W, so $a = 0$ is stationary. Differentiating again and using $f_W > 0$,

$$\frac{\partial^2 J_m}{\partial a^2}(a, L) = -\int_L^\infty m(r)\Big(f_W(a + r) + f_W(a - r)\Big)\,dr < 0$$

whenever $m > 0$ on a set of positive measure. Hence $J_m(\cdot, L)$ is strictly concave and $a = 0$ is its unique global maximiser. □

Proposition 3.3 establishes midpoint truthfulness as a dominant strategy: it holds for every admissible m, every L, and every realisation of $\mathcal{F}_i$. We restrict to $a = 0$, i.e. $\hat{\mu} = \mu_i$, for the remainder of the paper.

4 Optimal Reported Radius and Truthfulness of the Spread

4.1 Simplified Objective under Truthful Midpoint

Setting $a = 0$ gives $\Delta = -Z$ and $Y = W$, so (5) reduces to

$$g_0(r) = \mathbb{E}\Big[(r + Z)\,\mathbf{1}_{\{W \le -r\}} + (r - Z)\,\mathbf{1}_{\{W > r\}}\Big] \qquad r > 0 \tag{8}$$

The expected profit as a function of the reported radius alone is

$$H_m(L) := J_m(0, L) = \int_L^\infty m(r)\,g_0(r)\,dr \tag{9}$$

and the agent's problem reduces to $\sup_{L>0} H_m(L)$.

4.2 Structure of g_0 via Conditional Gaussian Regression

Lemma 4.1. *With $\kappa(n) = n/(n+1)$ from (1), for all $r > 0$,*

$$g_0(r) = r\,\mathbb{P}(|W| > r) + \kappa(n)\,\mathbb{E}\big[W\,\mathbf{1}_{\{W \le -r\}} - W\,\mathbf{1}_{\{W > r\}}\big] \tag{10}$$

and at $r = 0$,

$$g_0(0) = -\kappa(n)\,\mathbb{E}|W| < 0 \tag{11}$$

Proof. Since (Z, W) is jointly centred Gaussian (Remark 2.3), the Gaussian regression formula (Proposition A.8) gives

$$\mathbb{E}[Z \mid W] = \frac{\mathrm{Cov}(Z,W)}{\mathrm{Var}(W)}\,W = \frac{\sigma_i^2}{\sigma_i^2(n+1)/n}\,W = \kappa(n)\,W$$

Substituting into (8) and taking conditional expectations with respect to W:

$$\begin{aligned} g_0(r) &= \mathbb{E}\Big[(r + \kappa(n)W)\,\mathbf{1}_{\{W \le -r\}} + (r - \kappa(n)W)\,\mathbf{1}_{\{W > r\}}\Big] \\ &= r\,\mathbb{P}(|W| > r) + \kappa(n)\,\mathbb{E}\big[W\,\mathbf{1}_{\{W \le -r\}} - W\,\mathbf{1}_{\{W > r\}}\big] \end{aligned}$$

giving (10). At $r = 0$, Lemma A.9 gives $\mathbb{E}[W\,\mathbf{1}_{\{W \le 0\}}] = -\frac{1}{2}\mathbb{E}|W|$, so $g_0(0) = -\kappa(n)\,\mathbb{E}|W| < 0$. □

Lemma 4.1 shows that $g_0(0) < 0$ for all n: posting orders arbitrarily close to the midpoint is always unprofitable, as the agent is systematically adversely selected.

4.3 First-Order Condition for the Optimal Radius

Proposition 4.2. *Let m satisfy Assumption 2.4 and g_0 be continuous. At any $L > 0$ where m is also continuous,*

$$H_m'(L) = -m(L)\,g_0(L) \tag{12}$$

Any interior maximiser $L^* > 0$ with $m(L^*) > 0$ satisfies the first-order condition

$$g_0(L^*) = 0 \tag{13}$$

Proof. For $h > 0$,

$$\frac{H_m(L+h) - H_m(L)}{h} = -\frac{1}{h}\int_L^{L+h} m(r)\,g_0(r)\,dr \xrightarrow{h \to 0} -m(L)\,g_0(L)$$

by continuity of m and g_0 and the Lebesgue differentiation theorem (Theorem A.3). The left derivative is identical, giving (12). Setting $H_m'(L^*) = 0$ with $m(L^*) > 0$ yields (13). □

Remark 4.3. The condition (13) depends only on the joint distribution of (p^*, μ) through g_0, not on the shape of m. Any two strictly positive depth profiles give the same interior optimum L^*.

4.4 Incentive Compatibility in the Spread

Definition 4.4. The mechanism is *truthful in the spread* for agent i if the unique maximiser L^* of H_m satisfies $L^* = \sigma_i$.

Corollary 4.5. *Suppose g_0 is continuous, strictly increasing on $(0, \infty)$, and $\lim_{r\to\infty} g_0(r) \ge 0$. Then g_0 has a unique positive root $r^* > 0$, any interior maximiser of H_m with $m(r^*) > 0$ equals r^*, and truthful reporting $\hat{\sigma} = \sigma_i$ is optimal if and only if $r^* = \sigma_i$.*

Proof. By Lemma 4.1, $g_0(0) < 0$. Continuity and $\lim_{r\to\infty} g_0(r) \ge 0$ give a root $r^* > 0$ by the intermediate value theorem (Theorem A.4); strict monotonicity gives uniqueness. By Proposition 4.2, any interior maximiser with $m(r^*) > 0$ satisfies (13), hence equals r^*. □

5 The Depth Profile and Truthful Spread Reporting

The zero-crossing condition $g_0(\sigma_i) = 0$ is necessary for truthfulness but not sufficient: one must verify σ_i is a global maximiser, not just a stationary point. Since the root of g_0 is invariant to m (Remark 4.3), the depth profile must ensure H_m is single-peaked at that root.

Theorem 5.1. *Let Assumptions 1.3 and 2.4 hold, and suppose g_0 is continuous and strictly increasing on $(0, \infty)$ with unique root $r^* > 0$ and $g_0(0) < 0$. The following are equivalent:*

(i) $L^* = r^*$ is the unique global maximiser of H_m over $(0, \infty)$.

(ii) H_m is strictly increasing on $(0, r^*)$ and strictly decreasing on (r^*, ∞).

(iii) $m(r) > 0$ for all $r > 0$ and, for every continuity point $L > 0$ of m,

$$\operatorname{sgn}(H_m'(L)) = -\operatorname{sgn}(g_0(L)) \tag{14}$$

(iv) $m(r) > 0$ for all $r > 0$.

Moreover, when $m > 0$ everywhere, the mechanism is truthful in the spread for agent i if and only if $g_0(\sigma_i) = 0$.

Proof. **(iv)⇒(iii).** By (12), $\operatorname{sgn}(H_m'(L)) = -\operatorname{sgn}(g_0(L))$ when $m(L) > 0$. Since g_0 is strictly increasing with root r^*, we have $g_0(L) < 0$ on $(0, r^*)$ and $g_0(L) > 0$ on (r^*, ∞), so $H_m' > 0$ on $(0, r^*)$ and $H_m' < 0$ on (r^*, ∞).

(iii)⇒(ii). Immediate from the sign of H_m'.

(ii)⇒(i). A strictly single-peaked function has a unique global maximiser.

(i)⇒(iv). Suppose m vanishes on some open interval $(a, b) \subseteq (0, \infty)$. Then H_m is constant on (a, b), contradicting strict single-peakedness at r^* if $r^* \in (a, b)$. If $(a, b) \subset (0, r^*)$, then $H_m' = 0$ on (a, b) while $g_0 < 0$ there, so H_m is not strictly increasing on $(0, r^*)$, contradicting (ii). The case $(a, b) \subset (r^*, \infty)$ is analogous.

The final equivalence follows from Corollary 4.5. □

Remark 5.2 (Interpretation). Theorem 5.1 shows that the shape of m is irrelevant for spread truthfulness: any strictly positive m suffices. The binding constraint is a distributional calibration: the market size n must be set so that the unique root of g_0 equals σ_i. Since $\kappa(n) = n/(n+1)$ is agent-independent, this is a single calibration condition on n, not a separate condition per agent.

Corollary 5.3. *With $m > 0$ everywhere, the mechanism is truthful in the spread for agent i if and only if*

$$\sigma_i \, \mathbb{P}\big(|W| > \sigma_i\big) = \kappa(n) \, \mathbb{E}\big[W \, \mathbf{1}_{\{W > \sigma_i\}} - W \, \mathbf{1}_{\{W \le -\sigma_i\}}\big] \tag{15}$$

where $W \sim \mathcal{N}(0, \, \sigma_i^2 (n+1)/n)$.

Proof. The condition $g_0(\sigma_i) = 0$ in (10) reads

$$\sigma_i \, \mathbb{P}(|W| > \sigma_i) + \kappa(n) \, \mathbb{E}\big[W \, \mathbf{1}_{\{W \le -\sigma_i\}} - W \, \mathbf{1}_{\{W > \sigma_i\}}\big] = 0$$

which rearranges to (15). □

Remark 5.4. At first sight, condition (15) appears to depend on agent i's precision σ_i, suggesting that different agents require different market sizes for truthful reporting. This dependence is illusory, as the following standardisation shows. Write $W = \sigma_i\sqrt{(n+1)/n} \cdot \tilde{W}$ with $\tilde{W} \sim \mathcal{N}(0,1)$, so that $|W| > \sigma_i$ if and only if $|\tilde{W}| > \sqrt{n/(n+1)} =: \alpha(n)$. Substituting into (15) and dividing both sides by σ_i,

$$\alpha(n)\,\mathbb{P}\big(|\tilde{W}| > \alpha(n)\big) = \kappa(n) \cdot 2\,\mathbb{E}\big[\tilde{W}\,\mathbf{1}_{\{\tilde{W}>\alpha(n)\}}\big]$$

Every occurrence of σ_i has cancelled. Using $\kappa(n) = \alpha(n)^2$ and $\mathbb{P}(|\tilde{W}| > \alpha(n)) = 2(1-\Phi(\alpha(n)))$, this reduces to

$$\alpha(n)\,(1 - 2\Phi(\alpha(n))) = 2\phi(\alpha(n))$$

The cancellation is not accidental: the auction noise σ_i^2/n is proportional to σ_i^2, so a more uncertain agent perceives more noise in exactly the same proportion as their own signal uncertainty, leaving the signal-to-noise ratio $\kappa(n) = n/(n+1)$ agent-independent. Consequently, $n^\dagger$ is a universal constant that simultaneously implements truthful spread reporting for all agents.

6 Endogenous Entry and Market Liquidity

Sections 3–5 studied a fixed set of n agents. We now ask how the market populates itself when participation is voluntary. A pool of N potential entrants each compare their expected auction profit against an outside option.

6.1 Setup for the Entry Game

Assumption 6.1. There is a pool of $N \in \mathbb{N}$ potential entrants. Agent i's signal precision σ_i is drawn i.i.d. from a distribution F on $[\sigma_{\min}, \sigma_{\max}]$ with $0 < \sigma_{\min} < \sigma_{\max} < \infty$, where F is continuous and strictly increasing on its support.

Assumption 6.2. Each agent has access to an alternative trading venue yielding expected profit $c > 0$, independent of auction participation. An agent enters if and only if $H_m^{(n)}(\sigma_i) > c$, where $H_m^{(n)}(\sigma_i)$ denotes the maximised expected profit of agent i in a market with n participants.

Because $\kappa(n)$ is agent-independent, the optimal reported spread is proportional to the true spread for every agent.

Lemma 6.3. *Under Assumptions 1.3 and 2.4 with $m > 0$, the optimal reported spread for agent i in a market with n participants is*

$$r^*(n, \sigma_i) = \rho(n) \cdot \sigma_i \tag{16}$$

where $\rho(n) > 0$ depends only on n.

Proof. Since $W = \sigma_i\sqrt{(n+1)/n} \cdot \tilde{W}$ with $\tilde{W} \sim \mathcal{N}(0,1)$, the FOC $g_0(r^*) = 0$ in (10) becomes, after substitution:

$$r^*\,\mathbb{P}\left(|\tilde{W}| > \frac{r^*}{\sigma_i\sqrt{(n+1)/n}}\right) = \kappa(n) \cdot 2\sigma_i\sqrt{\tfrac{n+1}{n}} \cdot \mathbb{E}\left[\tilde{W}\,\mathbf{1}_{\left\{\tilde{W} > \frac{r^*}{\sigma_i\sqrt{(n+1)/n}}\right\}}\right]$$

Setting $u := r^*/(\sigma_i\sqrt{(n+1)/n})$, the equation depends on σ_i only through u, reducing to $h(u;\kappa(n)) = 0$ for a function h that does not involve σ_i. Let $u^*(n)$ be the unique positive root; then $r^*(n,\sigma_i) = u^*(n)\sigma_i\sqrt{(n+1)/n} =: \rho(n)\sigma_i$. □

6.2 Monotonicity of Expected Profit

Lemma 6.4. *Fix $n \geq 1$. The maximised expected profit $H_m^{(n)}(\sigma)$ is strictly decreasing in σ.*

Proof. By (9) at $L = r^*(n, \sigma) = \rho(n)\sigma$,

$$H_m^{(n)}(\sigma) = \int_{\rho(n)\sigma}^{\infty} m(r)\, g_0(r; \sigma, n)\, dr$$

Two effects operate when σ increases. First, the lower limit $\rho(n)\sigma$ shifts outward, shrinking the domain of profitable orders. Second, the variance $\sigma_i^2(n+1)/n$ of W increases, which one shows reduces $g_0(r; \sigma, n)$ pointwise for $r > \rho(n)\sigma$: since $\kappa(n)$ is independent of σ, the only dependence of g_0 on σ is through the scale of W, and the partial derivative $\partial g_0/\partial\sigma < 0$ follows from dominated convergence (Theorem A.2). Combining both effects gives $\partial H_m^{(n)}/\partial\sigma < 0$. □

Lemma 6.5. *Fix σ. The maximised expected profit $H_m^{(n)}(\sigma)$ is strictly decreasing in n.*

Proof. As n increases, $\kappa(n) = n/(n+1)$ increases toward 1, making $g_0(r; \sigma, n)$ more negative for small r and shifting the root $r^*(n, \sigma)$ outward. Both effects reduce $H_m^{(n)}(\sigma)$. Precisely: for $n' > n$, $\kappa(n') > \kappa(n)$ implies $g_0(r; \sigma, n') < g_0(r; \sigma, n)$ for all $r > 0$, and since $m \geq 0$,

$$\begin{aligned} H_m^{(n')}(\sigma) &= \int_{r^*(n',\sigma)}^{\infty} m(r)\, g_0(r; \sigma, n')\, dr \\ &< \int_{r^*(n,\sigma)}^{\infty} m(r)\, g_0(r; \sigma, n)\, dr = H_m^{(n)}(\sigma) \end{aligned}$$

using the pointwise ordering of g_0 and non-negativity of the integrand to the right of the root. □

Remark 6.6 (The two faces of competition). Lemma 6.5 reflects the tension identified in Remark 1.4: each additional participant improves price discovery (social benefit) but erodes informational rents (private cost). The profitable execution window, controlled by the root r^*, shrinks as $\kappa(n) \to 1$. It is this tension that generates a well-defined equilibrium participation level.

6.3 Rational Expectations Equilibrium (REE)

Definition 6.7. For $n \geq 1$, the *participation threshold* is

$$\sigma^*(n) := \sup\big\{\sigma \in [\sigma_{\min}, \sigma_{\max}] : H_m^{(n)}(\sigma) > c\big\}$$

The expected number of entrants from a pool of N is

$$\Psi_N(n) := N \cdot [1 - F(\sigma^*(n))]$$

A *rational expectations equilibrium* (REE) is a pair (n^*, σ^*) with $n^* = \Psi_N(n^*)$ and $\sigma^* = \sigma^*(n^*)$.

Theorem 6.8. *Let Assumptions 6.1 and 6.2 hold with $m > 0$ everywhere. Suppose $H_m^{(1)}(\sigma_{\min}) > c$ and $H_m^{(N)}(\sigma_{\max}) < c$. Then there exists a unique REE (n^*, σ^*) with $n^* \in \{1, \ldots, N\}$ and $\sigma^* \in (\sigma_{\min}, \sigma_{\max})$. At the REE, $H_m^{(n^*)}(\sigma^*) = c$, and the equilibrium is stable: $\Psi_N(n) > n$ for $n < n^*$ and $\Psi_N(n) < n$ for $n > n^*$.*

Proof. Define the excess entry function $\Phi_N(n) := \Psi_N(n) - n = N[1 - F(\sigma^*(n))] - n$.

Existence. As $n \to 0^+$, the noise $\sigma_i^2/n \to \infty$, making the auction trivially profitable, so $\sigma^*(0) = \sigma_{\max}$ and $\Phi_N(0) = N > 0$. At $n = N$, $H_m^{(N)}(\sigma_{\max}) < c$ gives $\Phi_N(N) < 0$. The intermediate value theorem (Theorem A.4) gives $n^* \in (0, N)$ with $\Phi_N(n^*) = 0$.

Uniqueness. By Lemma 6.5, $H_m^{(n)}(\sigma)$ is strictly decreasing in n, so the implicit function theorem (Theorem A.5) gives $d\sigma^*/dn > 0$, hence $1-F(\sigma^*(n))$ is strictly decreasing in n. Therefore $\Phi'_N(n) < -1 < 0$ for all n, giving at most one zero.

Stability. Follows from the strict monotonicity of Φ_N. □

6.4 The Truthful Market Size

Theorem 6.9. *Under Assumptions 1.3 and 2.4 with $m > 0$:*

(i) There exists a unique $n^\dagger \in (0, \infty)$ such that $\rho(n^\dagger) = 1$, given explicitly by

$$n^\dagger = \frac{(\alpha^*)^2}{1 - (\alpha^*)^2} \tag{17}$$

*where $\alpha^* \in (0, 1)$ is the unique solution to (**??**). At $n = n^\dagger$ all agents optimally report $\hat{\sigma}_i = \sigma_i$.*

(ii) For $n < n^\dagger$, $\rho(n) > 1$ (all agents overreport); for $n > n^\dagger$, $\rho(n) < 1$ (all agents underreport).

(iii) The distortion near $n^\dagger$ satisfies

$$|\rho(n) - 1| = \frac{1}{2\alpha^*(1 - (\alpha^*)^2)} \left| \frac{n - n^\dagger}{(n^\dagger)^2} \right| + O\Big((n - n^\dagger)^2\Big) \tag{18}$$

so the misreporting is small when n is close to $n^\dagger$.

(iv) The REE market size n^* equals $n^\dagger$ if and only if the outside option is set to

$$c^\dagger = H_m^{(n^\dagger)}\Big(\sigma^*(n^\dagger)\Big) \tag{19}$$

Proof. **(i) and (ii).** Since $\alpha(n) = \sqrt{n/(n+1)}$ is strictly increasing with $\alpha(n) \to 1$ and $\alpha^* < 1$, the equation $\alpha(n) = \alpha^*$ has a unique solution $n^\dagger = (\alpha^*)^2/(1 - (\alpha^*)^2)$. For $n < n^\dagger$, $\alpha(n) < \alpha^*$ so the root $r^*(n, \sigma_i)$ of g_0 satisfies $r^*(n, \sigma_i) > \sigma_i$, giving $\rho(n) > 1$. The reverse holds for $n > n^\dagger$.

(iii). The factor $\rho(n)$ is C^1 in $\alpha(n)$ near α^*. A first-order expansion of the root condition $h(u^*; \kappa(n)) = 0$ in $\alpha(n) - \alpha^*$, combined with $\alpha(n) = \sqrt{n/(n+1)}$ and the chain rule, gives (18).

(iv). By Theorem 6.8, the REE satisfies $H_m^{(n^*)}(\sigma^*(n^*)) = c$. Setting $n^* = n^\dagger$ uniquely determines $c = c^\dagger$ as in (19), and by uniqueness of the REE this is the only c that implements $n^* = n^\dagger$. □

Remark 6.10 (The liquidity–truthfulness tradeoff). Theorem 6.9 shows that the designer faces a genuine tradeoff only if the REE market size n^* (determined by c) differs from $n^\dagger$. Since depth $\mathcal{L} = 2Qn^*$ and the misreporting factor $|\rho(n^*) - 1|$ are both continuous functions of c, the designer can navigate between full truthfulness (at $c = c^\dagger$) and larger or smaller markets by adjusting c. If the designer can set $c = c^\dagger$ — for instance by subsidising or taxing participation — then maximum liquidity consistent with truthfulness and full truthfulness are achieved simultaneously.

6.5 Convergence of Liquidity as $N \to \infty$

Theorem 6.11. *Let the conditions of Theorem 6.8 hold for each $N \geq 1$, and let (n_N^*, σ_N^*) be the unique REE. As $N \to \infty$:*

(i) $n_N^* \to \infty$.

(ii) $\sigma_N^* \to \sigma_{\min}$.

(iii) If $F(\sigma_{\min}) = 0$ then $n_N^*/N \to 1$.

(iv) The aggregate spread $S_N := \min_{i:\, \sigma_i \leq \sigma_N^*} \sigma_i \to \sigma_{\min}$ almost surely.

(v) The aggregate depth $\mathcal{L}_N := 2Qn_N^* \to \infty$ almost surely.

(vi) The misreporting factor $\rho(n_N^*) \to \rho(\infty) < 1$ as $N \to \infty$. There exists a threshold pool size $N^\dagger$ at which $n_{N^\dagger}^* = n^\dagger$; for $N < N^\dagger$ agents overreport and for $N > N^\dagger$ they underreport.

Proof. **(i).** The condition $H_m^{(1)}(\sigma_{\min}) > c$ ensures $\Psi_N(n) > n$ for any fixed n when N is large enough, so $n_N^* > n$. Since n is arbitrary, $n_N^* \to \infty$.

(ii). By Lemma 6.5, $H_m^{(n)}(\sigma_{\min}) \to 0$ as $n \to \infty$. For large N (and hence large n_N^*), the indifference condition $H_m^{(n_N^*)}(\sigma_N^*) = c$ forces $\sigma_N^* \to \sigma_{\min}$.

(iii). Immediate from $1 - F(\sigma_N^*) \to 1 - F(\sigma_{\min}) = 1$.

(iv). The minimum $\sigma_{(1)}$ of n_N^* draws from $F|_{[\sigma_{\min}, \sigma_N^*]}$ satisfies

$$\mathbb{P}(\sigma_{(1)} > \sigma_{\min} + \varepsilon) \leq \left(1 - \frac{F(\sigma_{\min} + \varepsilon)}{F(\sigma_N^*)}\right)^{n_N^*}$$

which decays exponentially in n_N^*. The Borel–Cantelli lemma (Theorem A.7) gives $\sigma_{(1)} \to \sigma_{\min}$ almost surely.

(v). Each active agent contributes exactly Q to each side, so $\mathcal{L}_N = 2Qn_N^* \to \infty$ by part (i).

(vi). Since $n_N^* \to \infty$, $\alpha(n_N^*) = \sqrt{n_N^*/(n_N^* + 1)} \to 1 > \alpha^*$, so eventually $n_N^* > n^\dagger$ and the market is in the underreporting regime with $\rho(n_N^*) \to \rho(\infty) < 1$. For small N with $n_N^* < n^\dagger$ agents overreport; the market crosses the truthful point at $N = N^\dagger$. □

A Appendix

All results below are standard and are stated for completeness.

A.1 Integration and Differentiation

Theorem A.1 (Fubini–Tonelli)**.** *Let $(\Omega_k, \mathcal{A}_k, \mu_k)$, $k = 1, 2$, be σ-finite measure spaces and $h : \Omega_1 \times \Omega_2 \to \mathbb{R}$ measurable.*

(i) (Tonelli) *If $h \geq 0$ then*

$$\int_{\Omega_1 \times \Omega_2} h \, d(\mu_1 \otimes \mu_2) = \int_{\Omega_1} \left(\int_{\Omega_2} h \, d\mu_2 \right) d\mu_1$$

and both sides are equal (possibly $+\infty$).

(ii) (Fubini) *If $h \in L^1(\mu_1 \otimes \mu_2)$ then the same equality holds with all integrals finite.*

Applied in Section 2 to justify (3)*; integrability follows from Assumption 2.4 and $|g_a(r)| \leq C(1+r)$.*

Theorem A.2 (Dominated Convergence)**.** *Let (h_k) be measurable functions converging a.e. to h, with $|h_k| \leq g$ a.e. for some integrable $g \geq 0$. Then h is integrable and $\int h_k \, d\mu \to \int h \, d\mu$.*

Applied to justify: (a) differentiation of $g_a(r)$ under the expectation in Lemma 3.1, dominating function 1*; (b) differentiation of $J_m(a, L)$ under the integral in Proposition 3.3, dominating function $C\, m(r)(1 + r)$; (c) differentiation of $H_m^{(n)}(\sigma)$ with respect to σ in Lemma 6.4.*

Theorem A.3 (Lebesgue Differentiation Theorem)**.** *Let $h : \mathbb{R} \to \mathbb{R}$ be locally integrable. At any continuity point x of h,*

$$\lim_{\varepsilon \downarrow 0} \frac{1}{\varepsilon} \int_x^{x+\varepsilon} h(t) \, dt = h(x)$$

Applied in Proposition 4.2 to differentiate $H_m(L) = \int_L^\infty m(r) g_0(r) \, dr$.

A.2 Real Analysis

Theorem A.4 (Intermediate Value Theorem)**.** *If $h : [a, b] \to \mathbb{R}$ is continuous with $h(a)$ and $h(b)$ of opposite sign, then h has a zero in (a, b).*

Applied in Corollary 4.5 (root of g_0) and Theorem 6.8 (existence of the REE).

Theorem A.5 (Implicit Function Theorem)**.** *Let $G : \mathbb{R}^2 \to \mathbb{R}$ be C^1 near (x_0, y_0) with $G(x_0, y_0) = 0$ and $\partial_y G(x_0, y_0) \neq 0$. Then there exist neighbourhoods $U \ni x_0$, $V \ni y_0$ and a unique C^1 map $y : U \to V$ with $G(x, y(x)) = 0$ on U and*

$$y'(x) = -\frac{\partial_x G}{\partial_y G}\bigg|_{(x, y(x))}$$

Applied in Theorem 6.8 to differentiate $\sigma^*(n)$ and in Theorem 6.9 to compute the distortion near $n^\dagger$.

A.3 Probability Theory

Theorem A.6 (Strong Law of Large Numbers)**.** *If (X_i) are i.i.d. with $\mathbb{E}|X_1| < \infty$ then $n^{-1}\sum_{i=1}^{n} X_i \to \mathbb{E}[X_1]$ almost surely.*

Applied to conclude $\bar{\mu}_n \to \mu$ almost surely as $n_N^* \to \infty$.

Theorem A.7 (Borel–Cantelli Lemma)**.** *Let (A_k) be events. If $\sum_k \mathbb{P}(A_k) < \infty$ then $\mathbb{P}(A_k \text{ i.o.}) = 0$. If the events are independent and $\sum_k \mathbb{P}(A_k) = \infty$ then $\mathbb{P}(A_k \text{ i.o.}) = 1$.*

Applied in Theorem 6.11(iv) to show $\sigma_{(1)} \to \sigma_{\min}$ almost surely.

A.4 Gaussian Distributions

Proposition A.8 (Gaussian linear regression)**.** *If (U, V) is jointly centred Gaussian with $\mathrm{Var}(V) > 0$ then*

$$\mathbb{E}[U \mid V] = \frac{\mathrm{Cov}(U, V)}{\mathrm{Var}(V)}\, V \qquad a.s.$$

Applied in Lemma 4.1 with $U = Z$, $V = W$, giving $\mathbb{E}[Z \mid W] = \kappa(n)W$ where $\kappa(n) = n/(n+1)$.

Lemma A.9 (Half-moment of a symmetric variable)**.** *If $V \overset{d}{=} -V$ and $\mathbb{E}|V| < \infty$ then $\mathbb{E}[V\,\mathbf{1}_{\{V \le 0\}}] = -\mathbb{E}[V\,\mathbf{1}_{\{V > 0\}}] = -\frac{1}{2}\mathbb{E}|V|$.*

Proof. By symmetry, $\mathbb{E}[V\,\mathbf{1}_{\{V \le 0\}}] = -\mathbb{E}[V\,\mathbf{1}_{\{V > 0\}}]$. Since $\mathbb{E}|V| = \mathbb{E}[V\,\mathbf{1}_{\{V > 0\}}] - \mathbb{E}[V\,\mathbf{1}_{\{V < 0\}}] = 2\,\mathbb{E}[V\,\mathbf{1}_{\{V > 0\}}]$, the result follows. □

Applied in Lemma 4.1 to evaluate $g_0(0) = -\kappa(n)\,\mathbb{E}|W|$.

Proposition A.10 (Symmetry of the Gaussian CDF)**.** *If $V \sim \mathcal{N}(0, \sigma^2)$ then $F_V(-x) = 1 - F_V(x)$ for all $x \in \mathbb{R}$.*

Applied in Lemmas 3.2 and 3.3 to show the objective derivative vanishes at $a = 0$.